

ALAN ORDORICA

Software Engineer | Backend Systems | Cloud & Platform

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Summary

Multipotentialite¹ software engineer with a passion for learning and building across disciplines—quickly picking up tools from backend systems to embedded hardware and turning ambiguous challenges into deliverable solutions

Experience

Google

Sunnyvale, CA

Software Engineer | Google Cloud

May 2022 - Dec 2024

- Designed and implemented a high-fidelity control-plane simulator in C++/Python using gRPC and Protobuf, complete with pub/sub messaging and concurrency primitives (channels, callbacks). This simulator replicated complex Andromeda network behaviors, enabling developers to test new features in isolation and reducing full regression runtimes by **25%** (saving 1.5 SWE-years each year).
- Developed a multicast VPC-gateway emulation layer by extending the simulator to support 1,000+ gateways connected to a single controller. Created new C++ modules and Python orchestration scripts to emit and process messages, mocked missing data-plane components (GSLB, ESF port), and helped customers validate scalable pub/sub topologies—reducing customer onboarding time by **30%**.
- Replaced brittle monitoring hooks with modern observability: migrated from BNS watchers to Streamz/PK-based monitors. Wrote Python scripts to query Monarch for job health, parse Borg metadata, and auto-restart failed services; integrated these into the test harness, cutting device downtime by **27%** and eliminating false alarms.
- Built debugging and orchestration tools for isolated test sandboxes. Provisioned per-test sandboxes in Borg/Sandman, scripted metrics collection and auto-recovery, and authored log-parsing utilities that quickly pinpointed broken runfile paths or output misconfigurations—reducing release-blocking test failures by **30%** and speeding developer feedback loops by **40%**.
- Created pre-population strategies for Spanner test databases. Developed Python utilities to seed test DBs via textproto, Spanner backups, and Sandman snapshots; configured cron jobs and multi-sandbox topologies to ensure reproducible datasets and cut environment setup time significantly.
- Contributed to host-agent re-architecture, replacing an OpenFlow-based on-host agent with a new C++ On-Host Controller (OHC), integrating control-plane logic directly on VM hosts. Updated gRPC interfaces and host-path plumbing, improving reliability and deprecating legacy code.
- Authored design docs and diagrams for VPC-gateway and ESF-port integrations, weighing separate vs. co-located components and collaborating with cross-functional teams to finalize scalable architectures.

Apple

Sunnyvale, CA

Tools and Automation Engineer | CI Tools-Site Reliability

July 2020 - March 2021

- Standardized deployment of automated test racks by planning host topology, cabling, and rack configuration and documenting bring up and handoff procedures to streamline operations.
- Engineered a reproducible provisioning pipeline in Python, Bash, and Ansible to automate firmware flashing, disk imaging, credential setup, and MDM enrollment—ensuring consistent, hands off device bring up.
- Built SSH-based orchestration and monitoring utilities with parallel scripting and Splunk integration to automate fleet health checks, alerting, and imaging workflows—enabling rapid remediation from the terminal.

Full Stack Developer

Remote

Software Engineer

January 2020 - Present

- Developed modular React frontend and Node.js backend portfolio web apps with Docker Compose for local orchestration, enabling rapid environment setup.
- Developed Node.js RESTful microservices and ESP32-based automation projects using Python, Bash, and other tools, showcasing proficiency in hardware-software integration and system-level programming.

Technical Skills

Languages: C/C++, Python, JavaScript, Dart, Java, SQL, Bash, JSON, YAML

Libraries & Frameworks: Flutter, React, Next.js, Express.js, ESP-IDF, FreeRTOS

Technologies/Tools: Ansible, CMake, Git, GCP, Kubernetes, Docker, gRPC, Protobuf, CAD, Shaper3D, CI/CD, Microservices, Distributed Systems, ESP32, I²C, SPI, BLE, GDB

Education

¹ **Multipotentialite:** a person with multiple interests and an ability to learn quickly across different domains

